Satish Chandra Pamulapati

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Profile

Data Analyst with hands-on experience building detection and OCR pipelines, image restoration workflows, and vision-driven analysis systems. Comfortable taking a prototype to a reliable Pipeline: data prep, model training, evaluation, and integration code. Strong foundations in deep learning (CNNs, YOLO and similar tools) and practical CV tooling (OpenCV, PyTorch/TensorFlow).

Education

MSc, Machine Learning Systems and Control Sep 2025 - Present
Faculty of Engineering - Lund University, Sweden

- Conducted a Computer Vision workshop for ML students covering core image concepts, practical pipelines, and real-world use cases.
- Actively Working on Research for Computer Vision based 3d Keypoint Tracking and Flight 3D Path Estimation for Insects

Bachelor of Technology, Electronics and Communication Engineering
SRM Institute of Science and Technology, India

Skills

Core: Object Detection (YOLO,Resnet,VGG etc.,.), Image Classification, Image Segmentation, OCR Pipelines, Image Restoration, Image Analysis, 3D Computer Vision/Reconstruction
Methods: Transfer Learning, Fine-tuning, Hyperparameter Tuning, Feature Extraction, Evaluation (SSIM, MSE)
Tools: Python, NumPy, Pandas, OpenCV, PyTorch, TensorFlow, scikit-learn, Linux, Git
Platforms: Google Cloud Platform, Azure ML

Experience

Machine Learning Specialist (Contract), Kolomolo AB (Sweden) June 2026 - Present

- Working on Computer Vision and Machine Learning Projects

Associate Data Analyst / Machine Learning Intern, Zelar Sep 2024 - Aug 2025

- Built and improved OCR pipelines for 400-year-old historical documents, focusing on preprocessing and model-based text extraction for noisy, degraded scans.
- Created image restoration steps (denoising, deblurring, artifact cleanup) to boost OCR quality and reduce failure cases.
- Developed an image redaction pipeline to automatically detect and mask sensitive content in documents.
- Built a vision-based human movement analysis system using 3D pose classification for abnormal movement detection, with detection of behavior-related movement patterns.
- Delivered ML proof-of-concepts that helped convert **2 client pilots into long-term production contracts**, supported by clear Proof of Concepts.
- Built tools for advanced LLM-based summarization and structured extraction tasks when needed, mainly to make outputs easier for users to review and act on.

Projects

Multi-kilohertz Tracking of Insect Wings using Event-Based Cameras and Sensor Fusion Mar 2026 - Jun 2026

- Developed a event-based vision pipeline for high-frequency insect wingtip tracking using synchronized high-resolution cameras and an event camera.
- Synchronized five frame-based cameras using a microcontroller trigger and generated pseudo-ground-truth 2D wingtip labels using a pretrained SLEAP pose-estimation model.
- Temporally aligned event-camera data with frame-based keypoint annotations and trained a lightweight neural network to directly predict wingtip coordinates.
- Demonstrated recovery of high-frequency wing-motion patterns from event data, establishing a foundation for future low-latency, real-time closed-loop insect tracking.

Smart Attendance and Student Monitoring (YOLOv7) Dec 2022 - May 2023

- Trained and fine-tuned a YOLOv7-based detection pipeline for real-time classroom monitoring; led experiments to improve accuracy and stability and achieved around **85% Accuracy**.
- Implemented timestamp-based attendance logging to record entry and exit events and reduce manual interference.
- Added an activeness monitoring module to monitor inactivity and sleep-related behavior for classroom.
- Built a simple database integration to store inference outputs and attendance logs for reporting.

Deepfake Detection (Custom CNN) Jan 2023 - Apr 2023

- Designed and trained and optimized a custom CNN for deepfake classification; coordinated a 5-member team and structured experimentation.
- Achieved **88% accuracy** on real vs synthetic media and improved generalization through iterative feature extraction and tuning.

Image Restoration via Super Resolution Mar 2022 - Jun 2022

- Implemented super-resolution methods for restoring degraded images; compared results using **SSIM** and **MSE** with documented experiment tracking.

Publications (IEEE Xplore)

- An Intelligent Computer Vision based Application for Student Monitoring in Classroom
- Smart Farming Solutions for Crop Disease Prediction and Control using IoT and Machine Learning

Internships

AI Fellow, Metapix Jul 2023 - Apr 2024

- Worked on facial enhancement and gesture recognition using CNN-based feature extraction and transfer learning; supported real-time video processing workflows.

Amazon ML Summer School Jul 2022

- Completed an intensive ML program covering deep neural networks, probabilistic models, reinforcement learning, and practical experimentation.

Certifications

Meta Database Engineer (Coursera)
Stanford Algorithms Specialization
Harvard edX: Data Science and Machine Learning